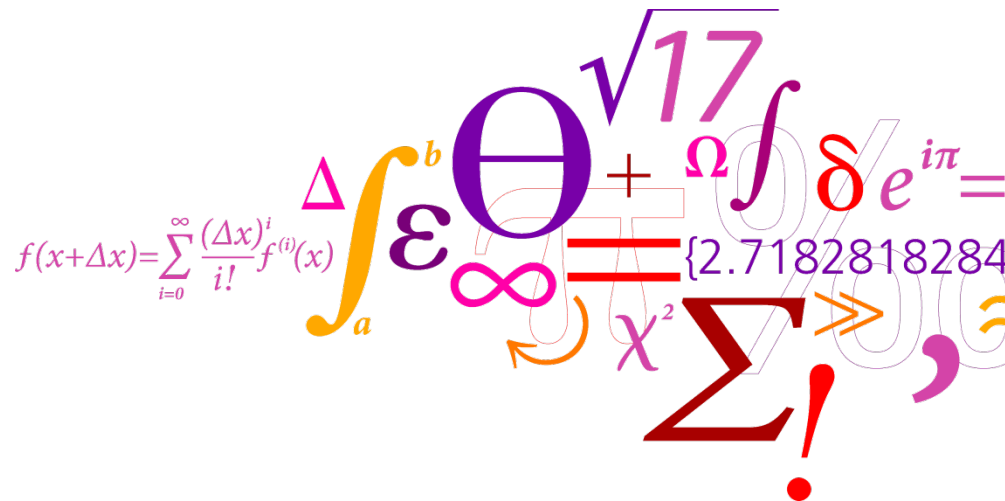


Gigabit Ethernet Switch

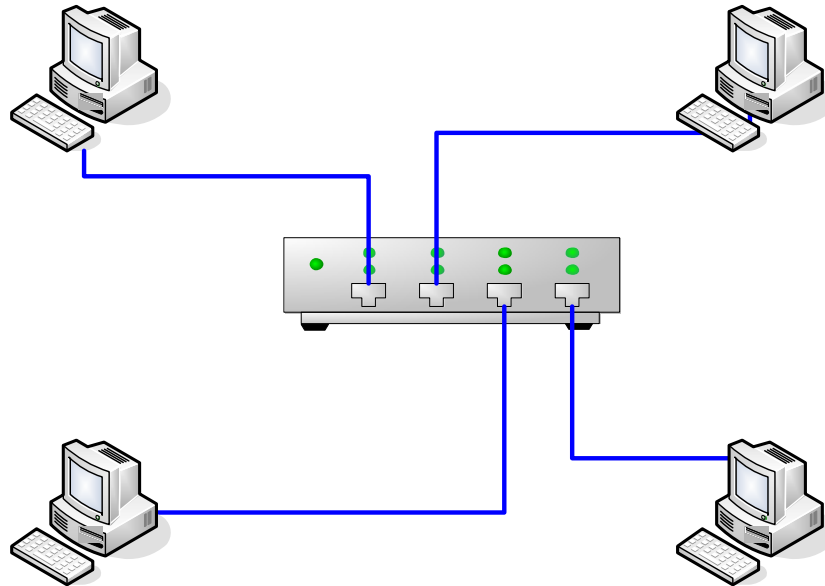
Project description

FPGA Design for Communications Systems

Layer 2



Motivation



- Ethernet is the protocol of choice for private and corporate Local Area Networks.
- Old shared bus (coaxial cables) and hub based solutions broadcast the traffic to all clients.
 - inefficient bandwidth utilization, high collision probability and limited security.
 - A switch will only send the traffic where it needs to go, and delete all invalid frames.
- Ethernet Local Area Networks are often used for file sharing, automatic hard drive image restoration etc
 - large bandwidth requirements -> this is why we need Gigabit Ethernet.

Overview of GE switch task

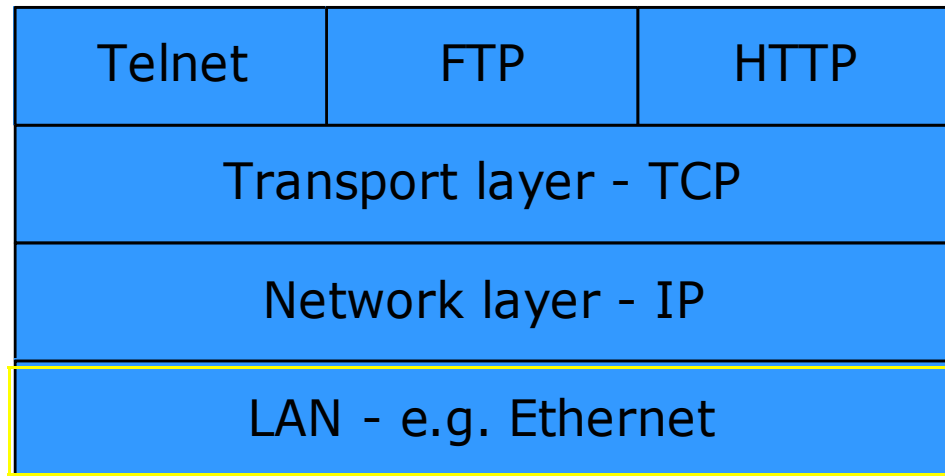
- 
- Objectives of the task
 - Basic switch functionality
 - Ethernet protocol and frame format
 - FCS-check (CRC-32)
 - MAC-learning
 - Fair queuing/scheduling
 - Architecture and buffering system
 - Transceiver interface
 - Design task
 - Implementation issues

Objectives of the task

- Design a 4-portet Gigabit Ethernet switch



Ethernet protocol and frame format



- Layer 2 of the ISO model – Data Link Layer.

Ethernet protocol and frame format

802.3 MAC Frame

Preamble	Start-of-Frame-Delimiter	Destination MAC	Source MAC	Ethertype/Length	Payload	FCS	Interframe gap
7 octets of 10101010	1 octet of 10101011	6 octets	6 octets	2 octets	46–1500 octets	4 octets	12 octets
		64–1518 octets					
72–1526 octets							

- Preamble and Interframe gap: Separates the MAC frames over the serial interface.
- SMAC and DMAC: Physical MAC addresses of the source and destination.
- Ether type: Specifies the payload type (ex. IP packet).
- Payload: Upper protocol data (IP packet, ARP request etc.)
- FCS: CRC-32 Frame Check Sum used for detecting corrupted frames.

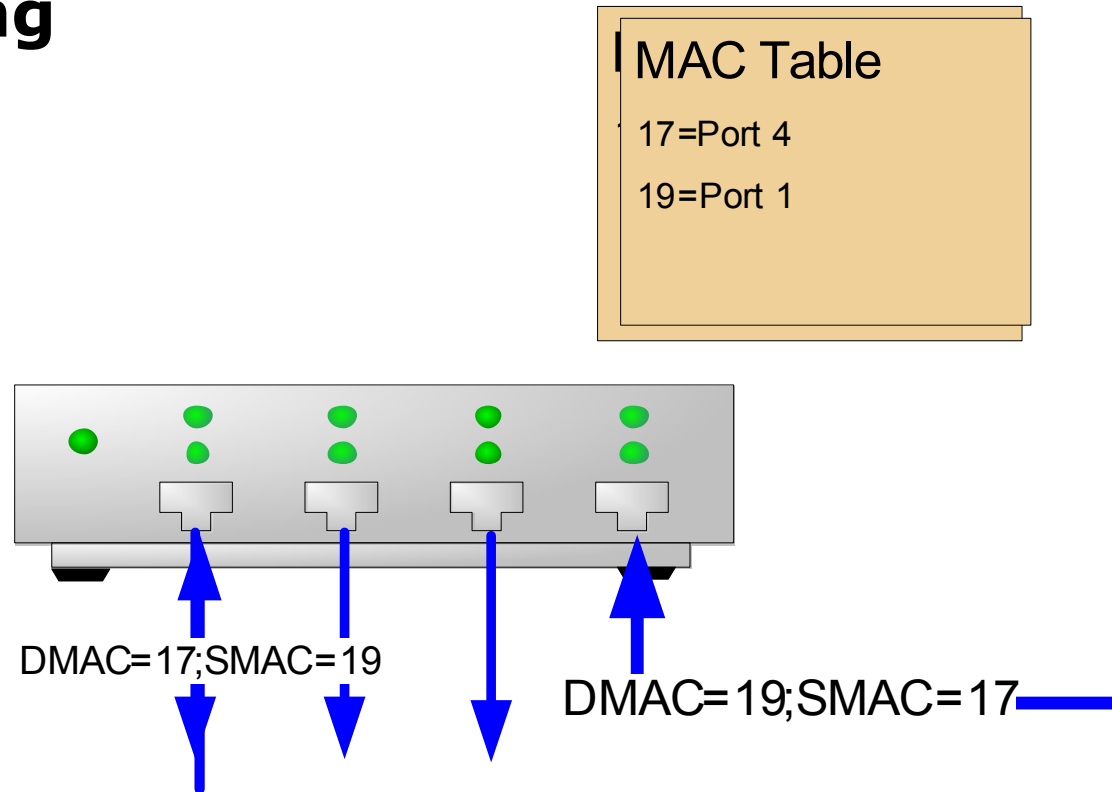
FCS Checksum

802.3 MAC Frame

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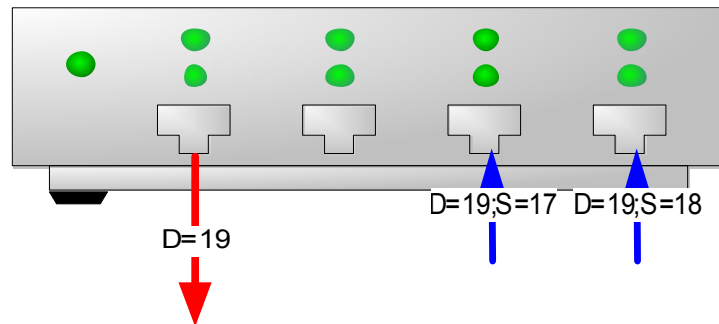
- 32-bit frame check sum
- Used to detect frames which have been corrupted during transmission
- Corrupted frames should not be forwarded
- You will implement the FCS calculation circuit in an upcoming exercise ☐

MAC-learning



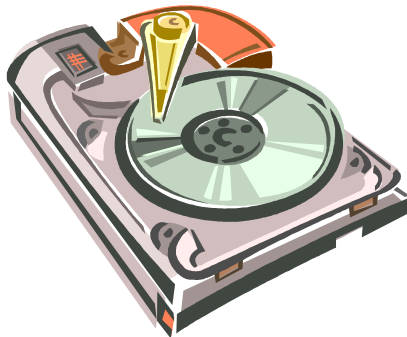
- MAC learning and MAC table lookup is not as simple as it may seem.

Fair queuing/scheduling



- Congestion caused by
 - inadequate switch capacity
 - multiple flows towards one host.
- Possible solution:
 - Frame buffering
 - Fair transmission scheduling

Switching - Considerations

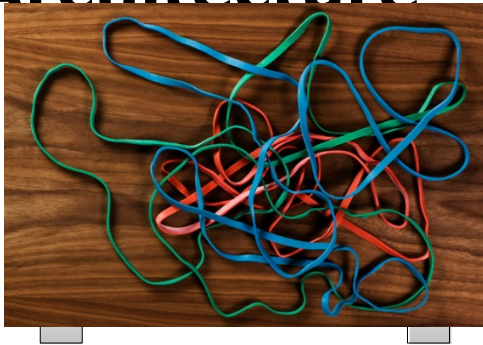


Buffering scheme

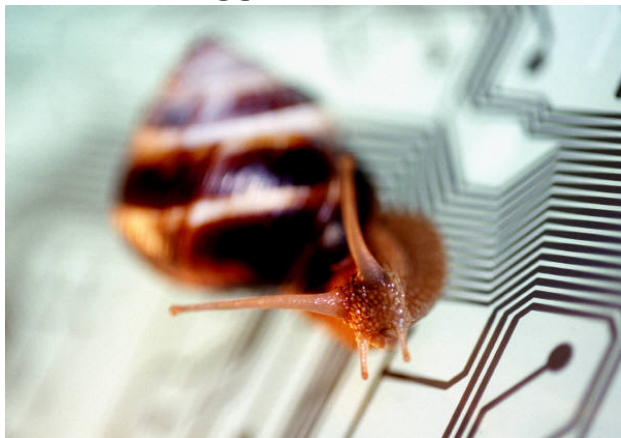


General switching architecture

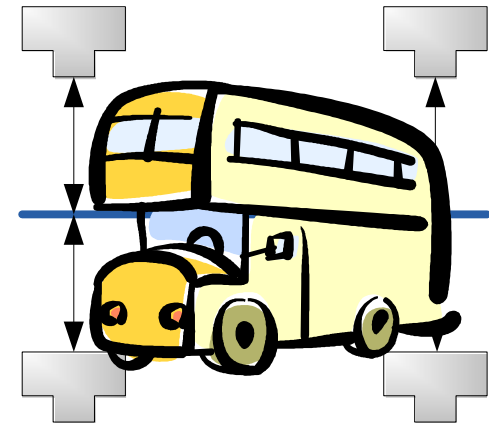
Architecture



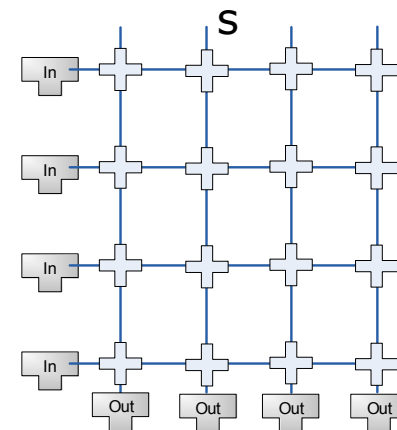
Full
Mesh



Shared
Memory



Bu



Cross-
bar

Buffering

- Input buffered
- Output buffered
- Virtual Output Queueing

Transceiver Interface

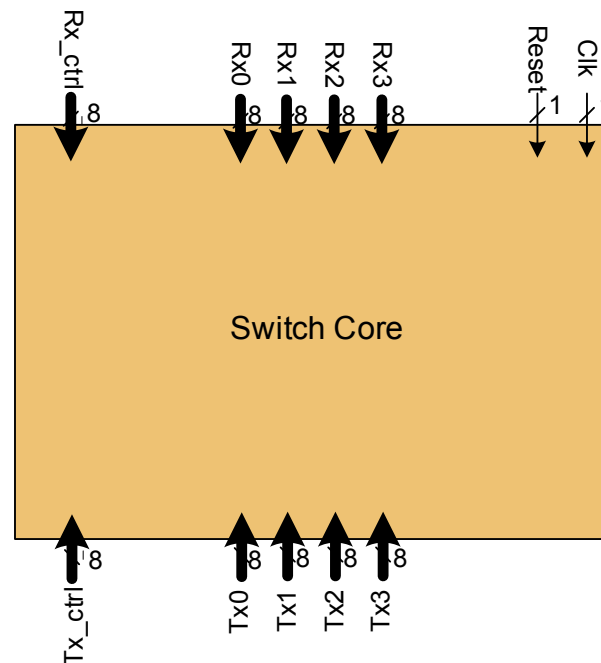
- The physical interface is beyond the scope of this project
- You need to supply a parallel interface which can later be connected to the physical layer.
- Gigabit Media Independent Interface (GMII) is appropriate.



Gigabit Switch Design Task

- Design a Gigabit Switch Core block in VHDL
- Networks interface runs at 1000Mb/s (8bit parallel)

$$= \frac{1000 \text{ Mb/s}}{8 \text{ bit}} = 125 \text{ MHz}$$



Design task (II)

- Requirements:
 - The design must be able to handle a total of 8k MAC addresses
 - The design must support frame sizes ranging from 64B to 1518B (excl. IFG and preamble).
 - GMII Interface
 - The switch must be non-blocking
 - No frame reordering of frames from the same flow
- Your job:
 - Decide a switch structure and consider the tradeoffs between different switch structures (buffering system/scheduler)
 - Design a testbench, which tests the implemented functionality

Design Issues

- Understand the function
 - Write a specification
 - Textual representation and block diagrams
 - Interfaces and functions within sub blocks
- Partition in smaller functional units
 - What can be done on chip
 - Data structures placed in external memory (and their size and BW)
- Testing
 - How can it be tested (and how much does the test cover???)

Team size: Appropriate for 3-4 persons